



National Institute of Technology Tiruchirappalli
Linear Algebra and Calculus (MAIR 12)

End Semester, Date: December 29, 2025

Timing: 9:30AM

Duration: 3 Hours

Max mark: 50

Do as directed

1. For each of the following statements, say whether it is true or false. If the statement is true, prove it. If false, justify why.

(a) The set of polynomials of degree ≤ 5 forms a vector space. (3)

(b) Suppose A is an 2×3 matrix, then $W = \{\vec{x} \in \mathbb{R}^3 : A\vec{x} = \vec{b}\}$ is a subspace when $\vec{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. (3)

(c) $\left\{ \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ -8 \\ 1 \end{bmatrix} \right\}$ is a basis of $\mathbb{R}^3$. (3)

2. Consider $\mathbb{R}^3$ with the following inner product

$$\langle \vec{u}, \vec{v} \rangle = \frac{1}{6}u_1v_1 + \frac{1}{8}u_2v_2 + \frac{1}{27}u_3v_3.$$

Let, $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$.

(a) Show that $\{\vec{v}_1, \vec{v}_2\}$ is an orthonormal set. (2)

(b) Find a vector $\vec{w} \neq 0$ such that $\langle \vec{w}, \vec{v}_1 \rangle = \langle \vec{w}, \vec{v}_2 \rangle = 0$. (3)

(c) For the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(\vec{x}) = (\langle \vec{x}, \vec{v}_1 \rangle, \langle \vec{x}, \vec{v}_2 \rangle, 0)$, determine $\text{Null}(T)$ and $\text{Range}(T)$ with their bases and dimensions, and verify the Rank-Nullity Theorem. (4)

3. For the quadratic form $Q(x, y, z) = 2x^2 + 2y^2 + z^2 + 2xy$, find the associated matrix, its eigenvalues and eigenvectors, determine the definiteness of Q , and reduce it to canonical form. (5)

4. Determine if the series $\sum_{n=1}^{\infty} \left(\frac{n-2}{n}\right)^n$ converges or diverges. (2)
5. For what values of p , the series $\sum_{n=2}^{\infty} (-1)^{n-1} \frac{(\ln n)^p}{n}$ convergent? (3)
6. If $a_n > 0$ and $\sum a_n$ converges, determine the convergence of $\sum \frac{a_n}{1+a_n}$. (2)
7. (a) If $x + y + z = u$, $y + z = uv$, $z = uvw$, find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$. (3)
- (b) Identify and discuss the stationary points of $f = x^3 + 12xy - 63(x + y) + y^3$ and hence find the extreme values, if any. (4)
- (c) Find the Taylor's series expansion of $f(x, y) = e^{xy}$ at $(1, 1)$ upto the third degree terms. (3)
8. (a) Transform to Cartesian and then evaluate $\int_0^{\pi} \int_0^a r^3 \sin \theta \cos \theta \, dr \, d\theta$. (3)
- (b) Evaluate $\iint (x^2 + y^2) \, dx \, dy$ where the region is the area of the parallelogram whose vertices are $(1, 0)$, $(3, 1)$, $(2, 2)$ and $(0, 1)$. (4)
- (c) Evaluate: $\int_0^{\log a} \int_0^x \int_0^{x+y} e^{x+y+z} \, dz \, dy \, dx$. (3)

Best wishes